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ELM-Enhanced PSO algorithm for Global Optimization

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Abstract—Particle Swarm Optimization (PSO) is a widely used metaheuristic algorithm for solving optimization problems. However, it often suffers from premature convergence and poor exploitation in complex search spaces. This paper proposes a novel three-stage hybrid optimization algorithm, termed ELM-PSO, which intelligently combines PSO and Extreme Learning Machine (ELM) to enhance the search capability for single-objective optimization problems. In the first stage, PSO is employed to explore the search space, and the optimization data are recorded for ELM training. In the second stage, both PSO and the trained ELM are used to generate new particles, with the data continually saved to refine the ELM. In the third stage, the PSO, ELM and a local search method are utilized to guide the search, as a particle has moved closer to regions of the optimum. Experimental evaluations on the CEC2014 benchmark functions demonstrate the effectiveness of the proposed method. The results show that the hybrid ELM-PSO algorithm outperforms state-of-the-art algorithms in terms of accuracy.

Index Terms—PSO, ELM, hybrid optimization, metaheuristic algorithm

I. INTRODUCTION

Single-objective optimization is a fundamental problem in many scientific and engineering domains. Over the years, numerous metaheuristic algorithms have been developed to tackle such problems, including Particle Swarm Optimization (PSO) [1], Differential Evolution (DE) [2], Cuckoo Search (CS) [3], Butterfly Optimization Algorithm (BOA) [4], and Dwarf Mongoose Optimization (DMO) [5]. Among these, PSO has gained popularity due to its simplicity and efficiency. However, PSO is prone to premature convergence and possesses limited search capabilities, which can result in slow convergence rates and low solution accuracy. These inherent limitations hinder its performance and motivate the development of enhanced PSO variants [15].

The performance of the Particle Swarm Optimization (PSO) algorithm is predominantly governed by three control parameters: the inertia weight, the cognitive acceleration coefficient, and the social acceleration coefficient. Early research often employed constant values for these parameters to ensure stable convergence. Chatterjee and Siarry [6] developed a nonlinear decreasing function for faster convergence. This line of inquiry has since expanded to include exponential and arbitrary adaptive functions, as surveyed in [7]. In contrast to the inertia weight, the cognitive and social acceleration coefficients initially received less attention and were typically set to constant values. However, the success of time-varying inertia weights prompted similar adaptations for acceleration coefficients [8]. Subsequent research has introduced nonlinear time-varying versions [9], strategies employing trigonometric functions [10], methods that utilize evaluation functions to dynamically adjust coefficients [11], and even approaches incorporating random values [12].

Beyond parameter tuning, significant research focuses on designing novel strategies and topological structures for PSO. The standard topology, where particles follow their personal best and the global best, often leads to premature convergence on multimodal problems, as the global best may be a local optimum. To address this, various advanced topologies have been proposed. One direction involves redefining the learning paradigm. The Prey-Predator PSO (PP-PSO) [13], improves convergence by identifying and transforming "slothful particles" (those with low velocities) and employs a proportional-integral controller to maintain population diversity. Multi-population and ensemble strategies have also proven effective. To mitigate oscillatory behaviors, the Two-Swarm Learning PSO (TSLPSO) [14] uses one subpopulation for dimensional-learning-based local search and another for comprehensive-learning-based global search.

Recent studies have enhanced PSO through self-adaptation

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and historical data reuse. [16] designed an archive-based PSO, where historically well-performed particles served as exemplars for updating the swarm. This concept was extended by [17] in the triple archive PSO (TAPSO), which categorized and preserved historical particles with different properties across multiple archives to better guide evolution. Furthermore, [18] effectively leveraged historical parameters and individuals in dynamic optimization, predicting the new optimum via rote learning and an FNN-based strategy trained on past data, demonstrating encouraging performance. Despite these efforts, most existing methods lack adaptive mechanisms that can effectively leverage historical search information during the optimization process.

In this paper, we present a novel PSO variant that incorporates an ELM to dynamically predict promising search directions. The key idea of ELM is that the input weights and biases of the hidden layer are randomly assigned and need not be tuned, while the output weights are analytically determined. This leads to extremely fast learning speed. During the search process, the ELM is trained on the historical positions of particles and their corresponding fitness values, allowing the algorithm to adjust the search strategy during different phases of the evolution. The proposed method is evaluated on the CEC2014 benchmark suite, and the results confirm its superiority over the standard PSO.

The remainder of this paper is structured as follows. Section II introduces ELM and PSO. Section III elaborates on the proposed ELM-PSO algorithm. Experimental results and comprehensive analyses are presented in Section IV. Finally, Section V concludes the paper and outlines directions for future research.

II. RELATED WORK

A. Extreme Learning Machine (ELM)

Extreme Learning Machine is a single-hidden-layer feedforward neural network proposed by Huang et al. [19]. Professor Huang Guangbin rigorously proved that if the activation function of the hidden layer is infinitely differentiable, the input weights and hidden layer biases of a single-hidden layer feedforward neural network can be randomly assigned. Once the input weights and hidden layer biases are randomly selected, the single-hidden layer feedforward neural network can be simply regarded as a linear system, and the output weights can be obtained through a generalized inverse matrix analysis of the hidden layer output matrix. Extreme Learning Machine is many times faster than traditional learning algorithms, including the backpropagation algorithm, and exhibits better generalization performance compared to conventional learning methods.

Given a set of training samples (x_i, t_i) where $x_i \in \mathbb{R}^n$ and $t_i \in \mathbb{R}^m$, the output of an ELM with L hidden nodes is:

$$f_L(x) = \sum_{i=1}^L \beta_i G(a_i, b_i, x) \quad (1)$$

where G is the activation function, a_i and b_i are the input weights and bias of the i -th hidden node, and β_i is the output weight. The goal is to minimize the error:

$$\min_{\beta} \|H\beta - T\| \quad (2)$$

where H is the hidden layer output matrix and T is the target matrix. The solution is given by:

$$\beta = H^{\dagger} T \quad (3)$$

where $H^{\dagger}$ is the Moore–Penrose generalized inverse of H .

B. Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is a population-based search optimization technique proposed by Eberhart and Kennedy, whose fundamental principle is inspired by the social behavior of bird flocks [1]. In the PSO algorithm, each particle in the population represents a potential solution to the optimization problem. Initially, particles with random velocities are randomly distributed in the search space. Then, the velocity of each particle is updated using its own historical best position (personal best experience), referred to as *pbest*, and the global best position of the entire population, referred to as *gbest*.

The velocity and position of each particle are updated as follows:

$$v_{ij}(t+1) = \omega v_{ij}(t) + c_1 r_{1i}(pbest_{ij} - x_{ij}(t)) + c_2 r_{2i}(gbest_j - x_{ij}(t)) \quad (4)$$

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t) \quad (5)$$

where ω is the inertia weight, j denotes a certain dimension of the variable, $j = 1, 2, \dots, n$, and n represents the dimensionality of the problem. i represents each particle in the population ($i = 1, 2, \dots, \mu$). x_{ij} and v_{ij} represent the position and velocity variables of the i -th particle, respectively. $pbest_i$ is the best position of the i -th particle, and $gbest$ is the best position of the entire population. c_1 and c_2 are acceleration coefficients. r_{1i} and r_{2i} are randomly generated numbers within the range $[0, 1]$. Thus, the PSO algorithm searches for the global optimum by simply adjusting the trajectory of each particle toward its own best position and the population's best position at each iteration.

A larger ω favors global exploration, while a smaller ω favors local exploitation. By linearly decreasing ω from a relatively large value to a small value during the optimization process, the algorithm can exhibit stronger global search capability in the early stages and stronger local search capability in the later stages. PSO can converge rapidly to the global optimum. However, as the value of ω decreases, PSO lacks global exploration ability in the later stages of the search, especially when solving multimodal problems, and may even fail to escape from local optima.

III. ELM-PSO ALGORITHM

The proposed hybrid ELM-PSO algorithm integrates an ELM into the PSO framework to improve its search capability. The optimization process is divided into three stages: the initial exploration stage, the intermediate hybrid stage, and the final exploitation stage. In the initial exploration stage, PSO is used to generate particles, and the historical data of each particle is recorded for subsequent use as training data for ELM. In the intermediate hybrid stage, ELM trained with the recorded historical data and PSO are probabilistically employed to generate new particles. In the final exploitation stage, the trained Extreme Learning Machine or local aggregation methods are used probabilistically to generate new particles, enabling better exploitation of the search space. The main steps of the algorithm are described below.

A. Initialization

The algorithm begins by initializing the particle positions and velocities within the search boundaries. The personal best and global best are initialized based on the fitness evaluation using the objective function. The ELM is initialized with the parameters of the number of hidden layers and the number of hidden layer nodes.

B. Iterative Process

The algorithm proceeds through three main stages:

1) *The initial exploration stage*: Standard PSO updates are performed without ELM intervention. Record the historical positions of all particles and the positions of the top-performing particles for subsequent use in training ELM.

2) *The intermediate hybrid stage*: An ELM model is trained using the historical positions of all particles and the positions of the top-performing particles. The input to the ELM is all particle positions in the population, and the output is the predicted superior position. Either PSO or the trained ELM is probabilistically selected to generate new particles. The update formula for particle positions via the trained ELM is:

$$x_i = gbest + rand() \times (predicted_pos - pbest_i) \quad (6)$$

where $gbest$ is the global best, $rand()$ is a random number within the range $[0, 1]$, $predicted_pos$ is the output of the ELM, $pbest_i$ is the personal best position.

3) *The final exploitation stage*: New particles are generated through the probabilistic use of ELM and the local aggregation strategy. The local aggregation strategy is applied:

$$a = \log((e - e^3) \times (i - 1)/(me - 1) + e^3) \quad (7)$$

$$pos = gbest + a \times (gbest - pbest) \quad (8)$$

Algorithm 1 ELM-Enhanced Particle Swarm Optimization (ELM-PSO)

Require: dimension D , population size ps , maximum generations me , The iteration number in the initial exploration stage $stage1$, iteration number in the intermediate hybrid stage $stage2$, ELM selection probability in the intermediate hybrid stage re , ELM selection probability in the final exploitation stage ret , local aggregation strategy selection probability in the final exploitation stage rlt

Ensure: Global best solution $gbest$, global best fitness $gbestval$

```

1: Initialize PSO velocities and positions
2: Initialize ELM parameters
3: Calculate fitness values
4: Update the optimal area  $gbest$ ,  $pbest$ 
5: Record all particles' positions
6: while  $i \leq me$  do
7:   Velocity Update by Eq.4
8:   Position Update by Eq.5
9:   if  $i > stage1$  and  $i < stage2$  then
10:     if  $rand() \leq re$  then
11:        $prepos \leftarrow ELM.predict(pos)$ 
12:       Position Update by Eq.6
13:     end if
14:   else if  $i \geq stage2$  then
15:     if  $rand() \leq ret$  then
16:        $prepos \leftarrow ELM.predict(pos)$ 
17:       Position Update by Eq.6
18:     else if  $rand() > 1 - rlt$  then
19:       Position Update by Eq.8
20:     end if
21:   end if
22:   fitness evaluation
23:   update Personal and Global Best
24:   Record all particles' positions
25:   ELM Retraining
26: end while
27: return  $gbest$ ,  $gbestval$ 

```

C. Pseudocode Explanation

The ELM-PSO algorithm integrates an ELM model into the standard PSO framework to improve search efficiency and solution quality. The algorithm operates in multiple stages, using the ELM model to predict promising particle positions and guide the swarm's movement.

Initialization Phase (Lines 1–5):

- Initialize the particle swarm with random positions and velocities within the search space.
- Initialize the parameters of the ELM model.
- Evaluate the fitness of each particle.
- Set the personal best position ($pbest$) for each particle and the global best position ($gbest$) for the swarm.
- Record all particles' positions for later ELM training.

Main Optimization Loop (Lines 6–26): The algorithm iter-

ates until the maximum generation count is reached. In each iteration:

- Update particle velocities and positions using standard PSO equations (Eq. 4 and 5).
- During the intermediate stage (between *stage1* and *stage2*), with a probability of *re*, the ELM model predicts a new position (*prepos*) for the particle, and the particle's position is updated using a dedicated rule (Eq. 6). Apply standard PSO updates particle velocities and positions with the remaining probability.
- In the final stage (after *stage2*), two mechanisms are applied:
 - With *ret* probability, the ELM-predicted position guides the update (Eq. 6).
 - With *rlt* probability, an alternative position update rule (Eq. 8) is applied.
 - With the remaining probability, standard PSO update rule is applied.
- Evaluate the fitness of all particles at their new positions.
- Update each particle's personal best (*pbest*) and the global best (*gbest*) if improved solutions are found.
- Record all particles' positions to expand the training dataset.
- Retrain the ELM model using the accumulated position-fitness data to improve its predictive accuracy for future iterations.

Output (Line 27): The algorithm returns the global best solution (*gbest*) and its corresponding fitness value (*gbestval*).

Key Features of ELM-PSO Algorithm:

- **Hybrid Strategy:** Combines swarm intelligence with machine learning (ELM) for adaptive search guidance.
- **Staged Evolution:** Applies different position update rules in different stages of the run to balance exploration and exploitation.
- **Online Learning:** The ELM model is periodically re-trained using data gathered during the search, allowing it to learn the landscape of the objective function.

IV. EXPERIMENTAL ANALYSIS

A. Experimental Setup

To comprehensively evaluate the performance of the proposed ELM-PSO algorithm, we conducted extensive experiments on the CEC 2014 benchmark suite [23], which consists of 30 complex real-parameter optimization functions. These functions include unimodal, simple multimodal, hybrid, and composition functions, presenting diverse challenges for optimization algorithms.

The CEC 2014 test suite comprises:

- **F1–F3:** Unimodal functions
- **F4–F16:** Simple multimodal functions
- **F17–F22:** Hybrid functions
- **F23–F30:** Composition functions

All functions are shifted and/or rotated to avoid biases toward specific search regions, with the global optimum located at

$o = [o_1, o_2, \dots, o_D]$ where o_i are randomly distributed in $[-80, 80]^D$.

For statistical robustness, each algorithm undergoes 30 independent runs on every function at both 10 and 30 dimensions. The computational budget is set to 100,000 function evaluations (FES) for 10-dimensional problems and 300,000 FES for 30-dimensional problems per run. Performance is assessed using the mean and standard deviation of solution errors across all runs.

B. Comparison Algorithms

The proposed ELM-PSO algorithm was compared against eight state-of-the-art and classic optimization algorithms:

- **Standard PSO (PSO)** [1]: The canonical Particle Swarm Optimization
- **Self-organizing hierarchical PSO (HPSO-TVAC)** [20]: A PSO variant with time-varying acceleration coefficients
- **Oscillatory PSO (OSC-PSO)** [21]: A PSO variant employing a strategy to force particles into oscillatory trajectories.
- **Nonlinear crossing strategy-based PSO (NCS-PSO)** [22]: A PSO variant with a nonlinear crossing strategy

The parameter settings of these algorithms are listed in Table I. Table II summarize the experimental results on the CEC 2014 benchmark suite for 30 dimensions. The reported error corresponds to the absolute difference between the best solution found over 30 independent runs and the true optimum of each objective function.

Non-parametric statistical tests are widely adopted for performance comparison among evolutionary algorithms [24]. In this study, the Wilcoxon signed-rank test is applied at a 5% significance level to evaluate the proposed ELM-PSO against several PSO variants. The algorithms are ranked based on their mean error values. For each algorithm, Table III lists the number of best ranking, the average ranking, and the outcomes of the statistical comparison in the last three rows using the symbols “+”, “=”, and “-”. Here, “+” indicates that ELM-PSO performs significantly better than the compared algorithm, “=” denotes statistically equivalent performance, and “-” signifies that ELM-PSO is significantly worse.

C. Experimental Results

1) **Overall Performance Comparison:** Table II presents the mean error values (function value minus optimum) and standard deviations obtained by all algorithms on the CEC 2014 benchmark functions. The best results for each function are highlighted in bold.

2) Algorithm Strengths and Weaknesses:

- 1) **ELM-PSO** demonstrated superior performance on multimodal and hybrid composition functions, particularly excelling on complex optimization problems (F1, F2, F6, F9, F11, F17, F18, F20, F21, F27, F28).
- 2) **OPSO** showed competitive performance on separable unimodal functions and some multimodal functions (F3, F4, F7, F8, F13, F14, F15, F24, F26).

TABLE I
ALGORITHM PARAMETER SETTINGS

Algorithms	Parameter Settings
PSO	Time-varying inertia weight: $w(t) = (1 - t/t_{\max})[w_s - w_e] + w_e$, $w_s = 0.9$, $w_e = 0.2$; constant acceleration coefficients, $c_1 = c_2 = 2.0$
HPSO-TVAC	Linear time-varying acceleration coefficients: $c_1(t) = \frac{t}{t_{\max}}[c_{1e} - c_{1s}] + c_{1s}$, $c_{1s} > c_{1e}$, $c_{1s} = 2.5$, $c_{1e} = 0.5$; $c_2(t) = \frac{t}{t_{\max}}[c_{2e} - c_{2s}] + c_{2s}$, $c_{2s} < c_{2e}$, $c_{2s} = 0.5$, $c_{2e} = 2.5$
OSC-PSO	$w_0 = w_{in} - (w_{in} - w_{end}) \cdot \frac{t}{t_{\max}}$ $w_{in} = 1.1$, $w_{end} = 0.1$;
NCS-PSOs	MDF: $e^{-t_{\max}/(t_{\max} - t)}$ $w_s = 0.99$, $w_e = 0.1$ $C_{lini} = C_{2e} = 3.5$, $c_{1e} = C_{2ini} = 0.0$ $t_{eross} = 205$
ELM-PSO	$stage1 = 0.2 * me$, $stage2 = 0.8 * me$, $re = 0.7$, $ret = 0.4$, $rlt = 0.3$

- 3) **PSO** performed well on specific function types, particularly achieving excellent results on F10, F16, F19, F22, F23, F25, F29, F30.
- 4) **NCS_PSO** showed limited competitiveness, with only 2 best performances on simpler functions (F5, F12).
- 5) **HPSO_TVAC** consistently underperformed across all function types, indicating that its time-varying acceleration coefficients may not be suitable for the CEC 2014 benchmark suite.

3) *Performance Consistency*: The standard deviation values indicate varying levels of solution stability across algorithms. ELM-PSO generally demonstrated reasonable stability, though some functions exhibited relatively high standard deviations (e.g., F1, F17, F21). OPSO showed particularly strong stability on several functions with low standard deviations (e.g., F5, F12, F13, F15, F26).

D. Experimental Conclusion

The experimental analysis on the CEC 2014 benchmark suite demonstrates that the proposed ELM-PSO algorithm significantly outperforms state-of-the-art optimization algorithms across diverse function types. The integration of ELM prediction with staged PSO evolution creates a powerful synergy that enables efficient exploration and precise exploitation of the search space. Statistical tests confirm the superiority of ELM-PSO. These results position ELM-PSO as a competitive alternative for complex real-parameter optimization problems.

V. CONCLUSION

This paper proposes a novel three-stage hybrid optimization algorithm, ELM-PSO, which intelligently integrates Particle Swarm Optimization with Extreme Learning Machine to enhance global optimization capabilities. The key innovation lies in the staged evolutionary framework that dynamically balances exploration and exploitation through machine learning guidance. The experimental results position ELM-PSO as a competitive and robust approach for solving complex real-parameter optimization problems.

Future work could investigate the scalability of these algorithms to higher dimensions and their performance on real-world optimization problems. Additionally, further analysis

of computational efficiency and parameter sensitivity would provide a more complete understanding of algorithm characteristics.

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TABLE II
MEAN ERROR AND STANDARD DEVIATION ON CEC 2014 FUNCTIONS (D=30)

Functions	Criteria	ELM-PSO	NCS_PSO	PSO	HPSO_TVAC	OPSO
F1	Mean	3.16E+06	4.98E+08	1.57E+07	7.58E+09	7.60E+06
	Std.	7.65E+06	4.19E+08	9.73E+06	1.57E+09	7.06E+06
F2	Mean	2.35E+02	4.57E+10	5.00E+03	1.63E+11	1.40E+03
	Std.	9.19E+03	9.16E+09	1.63E+03	7.69E+09	1.41E+03
F3	Mean	4.80E+03	2.44E+05	2.74E+03	1.74E+07	1.76E+03
	Std.	1.11E+03	9.21E+04	1.86E+03	2.26E+07	1.65E+03
F4	Mean	2.04E+02	4.73E+03	2.59E+02	6.08E+04	6.87E+01
	Std.	7.93E+00	1.21E+03	7.32E+01	2.75E+04	9.64E+01
F5	Mean	2.07E+01	2.04E+01	2.08E+01	2.15E+01	2.07E+01
	Std.	2.06E-02	4.20E-01	2.47E-02	4.42E-02	1.75E-02
F6	Mean	1.65E+01	2.74E+01	1.71E+01	5.37E+01	1.93E+01
	Std.	5.92E+00	3.66E+00	3.95E+00	7.01E-01	2.58E+00
F7	Mean	1.56E-02	4.73E+02	3.41E-02	1.70E+03	4.93E-03
	Std.	1.72E-01	6.52E+01	1.41E-02	2.66E+02	6.98E-03
F8	Mean	4.88E+01	2.64E+02	4.41E+01	5.93E+02	3.88E+01
	Std.	3.52E+00	5.77E+01	6.97E+00	1.50E+00	7.04E+00
F9	Mean	6.27E+01	2.69E+02	1.67E+02	6.72E+02	8.56E+01
	Std.	5.63E+00	2.25E-01	3.72E+01	1.69E+02	3.66E+01
F10	Mean	1.97E+03	4.61E+03	1.06E+03	1.01E+04	1.44E+03
	Std.	7.60E+02	6.17E+02	2.78E+02	4.56E+02	3.96E+02
F11	Mean	3.09E+03	5.15E+03	6.62E+03	1.01E+04	3.45E+03
	Std.	5.55E+02	3.55E+02	3.61E+02	7.98E+02	1.66E+02
F12	Mean	1.30E+00	9.67E-01	1.25E+00	1.11E+01	1.24E+00
	Std.	6.12E-02	2.62E-02	5.98E-01	9.87E-01	2.12E-01
F13	Mean	4.98E-01	6.06E+00	4.70E-01	1.34E+01	4.15E-01
	Std.	8.38E-02	3.46E-01	2.01E-01	6.46E-01	7.02E-02
F14	Mean	3.94E-01	1.54E+02	6.26E-01	6.54E+02	3.02E-01
	Std.	2.45E-03	3.11E+01	3.97E-01	5.35E+01	3.11E-02
F15	Mean	1.00E+01	3.26E+06	1.73E+01	1.58E+08	9.10E+00
	Std.	6.08E+00	1.40E+06	5.15E+00	1.35E+08	9.97E-01
F16	Mean	1.23E+01	1.24E+01	1.11E+01	1.45E+01	1.18E+01
	Std.	4.38E-02	4.86E-02	3.39E-01	2.51E-01	3.27E-01
F17	Mean	7.86E+05	5.51E+06	1.15E+06	1.08E+09	1.28E+06
	Std.	3.53E+05	4.79E+06	3.77E+05	8.17E+06	1.62E+06
F18	Mean	6.87E+03	2.72E+09	1.67E+06	1.93E+10	8.93E+03
	Std.	2.83E+04	3.84E+09	2.35E+06	2.21E+09	4.02E+03
F19	Mean	1.08E+01	2.09E+02	9.31E+00	3.91E+03	1.42E+01
	Std.	9.65E-01	1.48E+02	2.68E-01	1.75E+02	2.66E+00
F20	Mean	5.48E+02	1.96E+04	1.12E+03	2.48E+08	6.11E+02
	Std.	1.68E+02	1.35E+04	1.19E+02	2.41E+08	1.79E+02
F21	Mean	1.79E+05	3.09E+05	2.05E+05	1.20E+08	2.23E+05
	Std.	2.03E+05	6.07E+04	2.10E+04	4.86E+07	2.01E+05
F22	Mean	3.92E+02	4.45E+03	3.54E+02	8.74E+05	5.17E+02
	Std.	1.58E+02	4.67E+03	1.02E+02	1.12E+06	3.11E+02
F23	Mean	3.19E+02	6.41E+02	3.17E+02	7.15E+03	3.17E+02
	Std.	1.84E+00	2.87E+02	4.37E-01	8.61E+02	1.40E+00
F24	Mean	2.48E+02	3.93E+02	2.36E+02	6.28E+02	2.26E+02
	Std.	4.25E+00	3.96E+01	1.30E+01	1.04E+01	2.18E+00
F25	Mean	2.13E+02	2.35E+02	2.12E+02	6.49E+02	2.16E+02
	Std.	2.78E+00	8.01E-01	2.91E+00	2.30E-01	2.87E-01
F26	Mean	1.01E+02	1.07E+02	1.51E+02	6.19E+02	1.00E+02
	Std.	1.59E-01	1.23E+00	7.08E+01	6.68E+01	9.68E-02
F27	Mean	5.93E+02	1.03E+03	5.99E+02	1.99E+03	7.96E+02
	Std.	2.66E+02	3.35E+02	2.50E+02	3.94E+02	6.54E+00
F28	Mean	1.30E+03	1.43E+03	2.00E+03	1.08E+04	1.53E+03
	Std.	2.56E+02	5.04E+01	9.62E+02	8.95E+02	6.26E+02
F29	Mean	1.00E+04	1.48E+07	1.29E+03	1.07E+09	1.09E+07
	Std.	1.20E+04	2.08E+07	3.12E+01	1.24E+08	1.54E+07
F30	Mean	9.40E+03	2.35E+05	6.63E+03	2.70E+07	1.34E+04
	Std.	1.23E+04	2.55E+05	2.24E+03	1.47E+07	1.48E+04

TABLE III
WILCOXON SIGNED-RANK TEST ON CEC 2014 FUNCTIONS (D=30)

Functions	ELM-PSO	NCS_PSO	PSO	HPSO_TVAC	OPSO
F1	1	4(+)	3(+)	5(+)	2(+)
F2	1	4(+)	3(+)	5(+)	2(+)
F3	3	4(+)	2(-)	5(+)	1(-)
F4	2	4(+)	3(+)	5(+)	1(-)
F5	2	1(=)	4(+)	5(+)	3(=)
F6	1	4(+)	2(+)	5(+)	3(+)
F7	2	4(+)	3(+)	5(+)	1(-)
F8	3	4(+)	2(-)	5(+)	1(-)
F9	1	4(+)	3(+)	5(+)	2(+)
F10	3	4(+)	1(-)	5(+)	2(=)
F11	1	3(+)	4(+)	5(+)	2(+)
F12	4	1(-)	3(=)	5(+)	2(=)
F13	3	4(+)	2(=)	5(+)	1(-)
F14	2	4(+)	3(+)	5(+)	1(-)
F15	2	4(+)	3(+)	5(+)	1(-)
F16	3	4(=)	1(-)	5(+)	2(-)
F17	1	4(+)	2(+)	5(+)	3(+)
F18	1	4(+)	3(+)	5(+)	2(+)
F19	2	4(+)	1(=)	5(+)	3(+)
F20	1	4(+)	3(+)	5(+)	2(+)
F21	1	2(+)	3(+)	5(+)	4(+)
F22	2	4(+)	1(-)	5(+)	3(+)
F23	3	4(+)	1(=)	5(+)	2(=)
F24	3	4(+)	2(-)	5(+)	1(-)
F25	2	4(+)	1(-)	5(+)	3(+)
F26	2	3(=)	4(+)	5(+)	1(=)
F27	1	4(+)	2(=)	5(+)	3(+)
F28	1	2(+)	4(+)	5(+)	3(+)
F29	2	4(+)	1(-)	5(+)	3(+)
F30	2	4(+)	1(-)	5(+)	3(+)
Best ranking	11	2	8	0	9
Avg. ranking +/-/-	1.97	3.60 28/2/0	2.43 12/5/13	5.00 30/0/0	2.00 12/6/12

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